

Harsh Nishant Lalai

Incoming PhD Student at Johns Hopkins University

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About me

I'm interested in trustworthy AI, focusing on language models under uncertainty and multi-agent social interactions. My goal is to develop rigorous evaluations and robust methodologies that enable increasingly capable AI systems to be deployed safely and reliably in real-world settings.

Education

Aug 2026 – Present

Johns Hopkins University, Baltimore, MD

PhD in Computer Science

Advisor: Kristina Gligorić, Co-advisor: Benjamin Van Durme

Oct 2021 – Jul 2026

Birla Institute of Technology and Science, Pilani, Goa Campus, India

Bachelor of Engineering in Computer Science, Master of Science in Economics

GPA: 9.40 / 10

Undergraduate Thesis at Georgia Institute of Technology: *LLMs' Intermediate Thinking Tokens Are Weak Predictors of Human Cognitive Effort in Scientific Misconception Judgments*

Research Experience

Jul 2025 – May 2026

Georgia Institute of Technology, Atlanta, GA

Undergraduate Thesis Researcher

Mentors: Sashank Varma, Raj Sanjay Shah

Paper: *LLMs' Intermediate Thinking Tokens Are Weak Predictors of Human Cognitive Effort in Scientific Misconception Judgments*

Investigating whether LLM reasoning-effort signals (thinking tokens) correspond to human cognitive effort on scientific misconceptions.

[Submitted to ACL Rolling Review \(May 2026\).](#)

Jun 2025 – Mar 2026

Harvard University, Cambridge, MA

Research Collaborator, Visual Computing Group

Mentors: Hanspeter Pfister, Grace Guo

Paper: *When Visuals Aren't the Problem: Evaluating Vision-Language Models on Misleading Data Visualizations*

Studying VLM robustness to misleading chart and caption pairs and the ability to detect fine-grained reasoning and visualization errors.

[Submitted to ACL Rolling Review \(March 2026\).](#)

May 2024 – Mar 2025

Emory University / Brock University, Atlanta, GA

Visiting Scholar

Mentors: Ali Emami, Yi-Chia Wang

Paper: *The World According to LLMs: How Geographic Origin Influences LLMs' Entity Deduction Capabilities*

Evaluating implicit geographic bias in LLMs through a multi-turn entity deduction game. Used notable people and culturally significant objects from diverse regions as entities, and investigated the effect of pre-training frequency and popularity on deduction.

[Presented at COLM 2025.](#)

Dec 2023 – May 2024	Pennsylvania State University , University Park, PA <i>Research Intern, PIKE Group</i> Mentor: Dongwon Lee Paper: <i>From Intentions to Techniques: A Comprehensive Taxonomy and Challenges in Text Watermarking for Large Language Models</i> Analyzed text-watermarking research by intentions, evaluation datasets, and addition / removal methods to construct a unified taxonomy, and highlighted gaps and open challenges in protecting text authorship. Presented at NAACL 2025 Findings.
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Employment

Jan 2026 – Jun 2026	SingleStore , Hyderabad, India <i>Software Engineering Intern, AI Team</i> Building an experimental semantic retrieval benchmark for SingleStoreDB, comparing transformer-based encoders (e.g., MiniLM, E5, Cohere Embeddings) and vector-index configurations.
May 2023 – Jul 2023	Jio Brain (formerly Jio Platforms) , Mumbai, India <i>Software Engineering Intern, Automated Transcription Team</i> Developed preprocessing utilities and evaluation scripts for JioBrain’s Speech-to-Text module, automating dataset ingestion and inference-quality testing across multilingual pilot datasets for Hindi, Gujarati, Marwari, Marathi, and Tamil.

Honors and Awards

2025	K. M. Birla Foundation Student Travel Grant Awarded for travel to COLM 2025. Given to selected students with an accepted first-author paper at a top-tier international conference outside India.
2024	MITACS Globalink Research Internship Scholarship Selected from over 30,000 applicants for top Canadian research universities. Fully funded research internship at Brock University under Professor Ali Emami.
2023 – 2026	Merit Scholarship, BITS Pilani Awarded for six consecutive semesters to the top 1% of approximately 800 students for academic performance.

Publications

2026	1. Lalai, H. N. , Shah, R. S., Pfister, H., Varma, S. & Guo, G. When Visuals Aren’t the Problem: Evaluating Vision-Language Models on Misleading Data Visualizations. <i>arXiv preprint</i> (2026).
2025	2. Lalai, H. N. , Ramakrishnan, A. A., Shah, R. S. & Lee, D. <i>From Intentions to Techniques: A Comprehensive Taxonomy and Challenges in Text Watermarking for Large Language Models</i> in <i>Findings of the Association for Computational Linguistics: NAACL</i> (2025). 3. Lalai, H. N. et al. The World According to LLMs: How Geographic Origin Influences LLMs’ Entity Deduction Capabilities. <i>Second Conference on Language Modeling (COLM)</i> (2025).

Open Source Contributions

I believe in open-source development for reproducibility and impact.

MisVisBench Dataset

MisVisBench is a dataset of carefully curated visualizations spanning diverse visualization and reasoning errors, enabling systematic assessment of model robustness to deceptive visual communication and analytical mistakes.

Geo2oQ+ Dataset

Geo2oQ+ is a geographically balanced dataset designed to evaluate implicit geographic biases in LLMs through a multi-turn deduction game inspired by 20 Questions. It includes entities of two types, Notable People and Culturally Significant Things, from diverse regions across the globe.

TwentyQ: The 20 Questions Game Engine (Python Package)

A pip package that lets users simulate and evaluate a 20 Questions-style guessing game using LLMs such as OpenAI GPT, Anthropic Claude, Google Gemini, and Huggingface-hosted models.

A Taxonomy of Text Watermarking for LLMs (NAACL 2025)

An open-source, continuously updated repository that organizes the text-watermarking literature by intentions, methods, and evaluation datasets, serving as a community reference for researchers and practitioners.

Teaching

Fall 2024	Undergraduate Teaching Assistant <i>Birla Institute of Technology and Science, Pilani, Goa Campus, India</i> International Economics (ECON F311). Created a term-paper-based assignment to give students hands-on research exposure.
Spring 2024	Undergraduate Teaching Assistant <i>Birla Institute of Technology and Science, Pilani, Goa Campus, India</i> Econometric Methods (ECON F241). Designed coding exercises to give students hands-on experience with R and Stata.
Fall 2023	Undergraduate Teaching Assistant <i>Birla Institute of Technology and Science, Pilani, Goa Campus, India</i> Financial Management (ECON F315). Created homework, assignments, and assisted students with their capstone projects.
Spring 2023	Undergraduate Teaching Assistant <i>Birla Institute of Technology and Science, Pilani, Goa Campus, India</i> Principles of Economics (ECON F211). Created homework and assignments for the course.

Service

Reviewing: ACL Rolling Review (ARR) 2026, 2025; Conference on Language Modeling (COLM) 2026; International Conference on Mathematics and Computing (ICMC) 2023.

Volunteering: International Conference on Mathematics and Computing (ICMC) 2023; International Conference on Data-Driven Computing and Intelligent Systems 2023; International Conference on Advances in Data-Driven Computing and Intelligent Systems 2022.

Technical Skills

Proficient: Python, C, C++, PyTorch, TensorFlow, Scikit-learn, Pandas

Comfortable: Java, $\LaTeX$, Git, NLTK

Familiar: R, MySQL, HTML, CUDA

Languages

Native Proficiency: English, Hindi, Gujarati, Kutchi

Expert Proficiency: Marathi

Hobbies

Sports & Fitness: Football, Cricket, Mixed Martial Arts, Gym Training, Badminton, Table Tennis

Reading: Autobiographies

Media & Entertainment: Sports, Sports Documentaries, Action & Thriller Movies and TV Series, e-Sports

References

Dr. Kristina Gligorić, Assistant Professor

Department of Computer Science, *Johns Hopkins University*

Dr. Benjamin Van Durme, Professor

Department of Computer Science, *Johns Hopkins University*

Dr. Sashank Varma, Professor

School of Interactive Computing, *Georgia Institute of Technology*

Dr. Yi-Chia Wang, Principal Research Scientist

Carnegie Mellon University. Visiting Scholar, *Stanford University*